

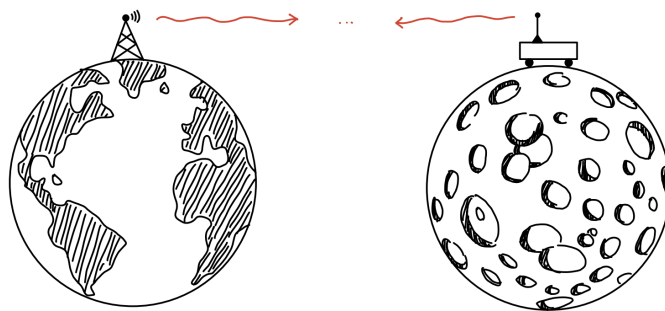
Exercise 12 - Time Delays

1 Time Delays

Until now, we have only considered systems without time delays. In other words, we have only considered control systems in which inputs, disturbances and references affect the output instantaneously.

However, in many practical system, and thus in the real world, systems frequently exhibit time delays originating from various sources, as sensing, computation, communication, etc.

A simple example to gain some intuition could be a lunar rover controlled from Earth: every command takes a few seconds to reach the rover, and the sensor data take time to come back. Because of this delay, the rover reacts late, and if the delay is large, its movements can become oscillatory or hard to control.



But how can we model time delays? How can we take them into account when designing controllers?

We can represent a time delay as an operator that shifts an input signal $u(t)$ by a time $T \geq 0$. This would result in an output signal defined as

$$y(t) = u(t - T)$$

A very important characteristic of a time delay is that it is a linear operator. Thus, time delays fit well within the framework of LTI systems and this means that we can continue to use the tools we have learned so far to analyze them.

We can also compute the transfer function of a time delay.

To do so, we consider a general input $u(t) = e^{st}$. If we now apply a time delay of T to this input, we obtain the following output

$$y(t) = u(t - T) = e^{s(t-T)} = e^{-sT} e^{st} = e^{-sT} u(t)$$

Thus, the transfer function of a time delay is given by

$$G(s) = e^{-sT}$$

We can easily see that its transfer function is **not rational**. This prevents us from using some tools as the Root Locus method, which require rational transfer functions. For this reason, frequency domain methods are often preferred when dealing with time delays.

Recall that in the frequency domain, we substitute $s = j\omega$. Thus, the frequency response of a time delay is given by

$$G(j\omega) = e^{-j\omega T}$$

This expression has some interesting properties:

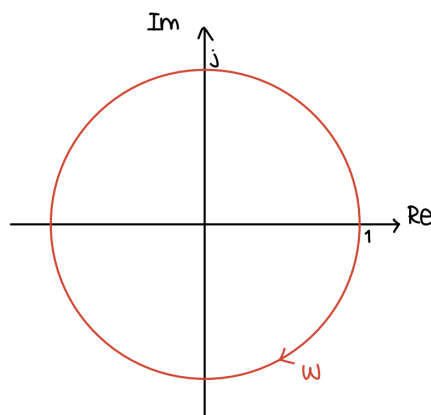
$$|G(j\omega)| = |e^{-j\omega T}| = 1$$

$$\angle G(j\omega) = \angle e^{-j\omega T} = -\omega T$$

This can be summarized as:

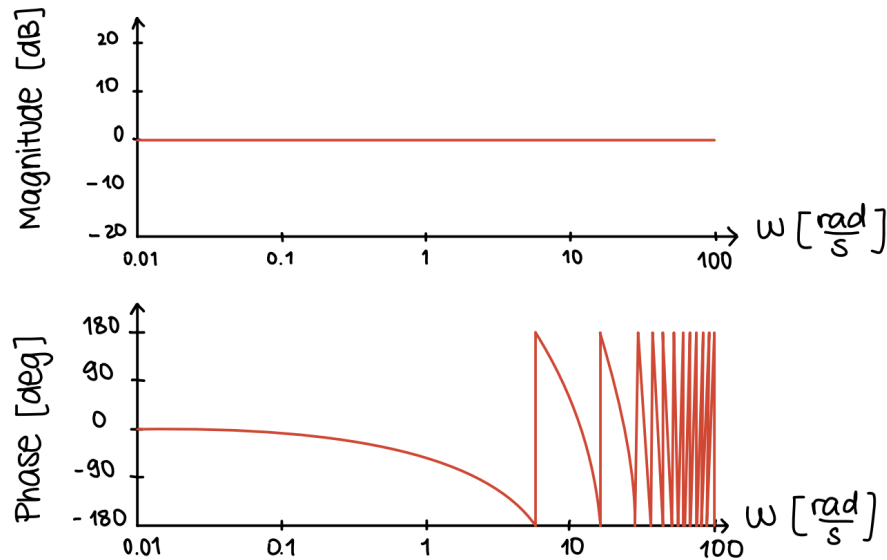
- Its magnitude is equal to 1 for all frequencies ω , meaning that a time delay does not attenuate or amplify signals at any frequency.
- Its phase is equal to $-\omega T$, meaning that a time delay introduces a phase lag that increases linearly with frequency.

These two characteristics can also be visualized in a polar plot



In fact, we see that the time delay represents a point moving around the unit circle in the clockwise direction as the frequency ω increases.

We can also visualize the effect of a time delay in a Bode plot



Note: The magnitude plot is a horizontal line at 0 dB (1 in linear scale), while the phase plot is wrapped between $(-\pi, \pi]$ to avoid unbounded values.

We can summarize everything by saying that since a time delay does not affect the magnitude of the frequency response, we only have to be very careful about its phase effects.

1.1 Phase margin

Recall from last week that the phase margin is defined as

$$\varphi_{m,0} = 180^\circ + \angle L(j\omega_{gc})$$

where ω_{gc} is the gain crossover frequency, i.e., the frequency at which the magnitude of the open-loop transfer function $L(j\omega)$ is equal to 1 (0 dB).

Now, if we introduce a time delay of T seconds in the system, our "new" phase becomes

$$\angle L_{new}(j\omega_{gc}) = \angle L(j\omega_{gc}) - \omega_{gc}T$$

This means that the new phase margin is given by

$$\varphi_{m,T} = 180^\circ + \angle L_{new}(j\omega_{gc}) = 180^\circ + \angle L(j\omega_{gc}) - \omega_{gc}T = \varphi_{m,0} - \omega_{gc}T$$

This shows that the presence of a time delay reduces the phase margin of the system by an amount equal to $\omega_{gc}T$.

If the delay is too large, or if the gain crossover frequency is too high, the phase margin can become negative, leading to instability in the closed-loop system.

1.2 Control design in presence of time delays

One way to design controllers in presence of time delays is given by the following steps:

1. Design the feedback controller assuming zero time delay
2. Evaluate the effective phase margin when the time delay is considered $\varphi_{m,T} = \varphi_{m,0} - \omega_{gc}T$. If the phase margin is too low or negative:
 - Reduce the gain crossover frequency ω_{gc} .
 - Increase the phase margin $\varphi_{m,0}$ of the system without delay.
3. Iterate until a good performance is achieved.

1.3 Approximation of time delays in rational functions

As mentioned before, the transfer function of a time delay is not rational, which prevents us from using some tools.

However, we can approximate a time delay with a rational transfer function using two main methods.

Taylor series approximation

One way to approximate a time delay is to expand its transfer function in a Taylor series around $s = 0$. The Taylor series expansion of e^{-sT} is given by

$$e^{-sT} = 1 - sT + \frac{(sT)^2}{2!} - \frac{(sT)^3}{3!} + \dots$$

If we truncate this series and for example keep only the first three terms, we obtain the following first-order approximation of the time delay:

$$e^{-sT} \approx 1 - sT + \frac{1}{2}(sT)^2$$

However, this approximation has some limitations. In particular, it does not preserve the magnitude characteristic of the original time delay. In fact, for high frequencies ω , the magnitude of the Taylor series approximation will diverge from 1, which is not the case for the actual time delay.

For this reason, the Taylor series approximation is not often used in practice, especially for high-frequency applications.

Padé Approximation

A more accurate method to approximate a time delay is given by the Padé approximation. The Padé approximation is a rational function that approximates a given function by matching its Taylor series expansion up to a certain order.

For simplicity, we will only consider the first-order Padé approximation of a time delay, which is given by

$$e^{-sT} \approx k \frac{s+p}{s+q}$$

where k , p and q are constants. By matching the Taylor series expansion, we can determine these constants as

$$k \frac{s+p}{s+q} \approx 1 - sT + \frac{1}{2}(sT)^2$$

If we solve this equation, we find that

$$ks + kp = s + q - s^2T - sqT + \frac{1}{2}s^3T^2 + \frac{1}{2}qs^2T^2$$

By comparing everything we get

$$k = 1, \quad p = -\frac{2}{T}, \quad q = \frac{2}{T}$$

Thus, the first-order Padé approximation of a time delay is given by

$$e^{-sT} \approx \frac{\frac{2}{T} - s}{\frac{2}{T} + s}$$

Concluding, with this approximation, we are able to use tools such as the Root Locus method again.

Note that higher-order Padé approximations can also be used for more accuracy, but they result in more complex rational functions.

2 Non-linear Systems

So far, we have learned and used many tools that are only valid for linear time-invariant (LTI) systems. However, most real-world systems are non-linear, meaning that their behavior cannot be accurately described by linear equations.

In Exercise 3, we have already seen an example of a non-linear system, that is the pendulum. Furthermore, we saw that to bridge the gap between non-linear and linear systems, we can linearize the non-linear system around an equilibrium point using the Jacobian.

Let's revise this concept again-

Consider a system described state-space equations:

$$\dot{x}(t) = f(x(t), u(t))$$

$$y(t) = h(x(t), u(t))$$

We can linearize this system around an equilibrium point (x_e, u_e) by computing the Jacobian matrices of f and h with respect to x and u . The linearized system is then given by the following matrices:

$$A = \left. \frac{\partial f(x, u)}{\partial x} \right|_{(x_e, u_e)} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}_{(x_e, u_e)} \in \mathbb{R}^{n \times n}$$

$$B = \left. \frac{\partial f(x, u)}{\partial u} \right|_{(x_e, u_e)} = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \dots & \frac{\partial f_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \dots & \frac{\partial f_n}{\partial u_m} \end{bmatrix}_{(x_e, u_e)} \in \mathbb{R}^{n \times m}$$

$$C = \left. \frac{\partial h(x, u)}{\partial x} \right|_{(x_e, u_e)} = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \dots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial x_1} & \dots & \frac{\partial h_p}{\partial x_n} \end{bmatrix}_{(x_e, u_e)} \in \mathbb{R}^{p \times n}$$

$$D = \left. \frac{\partial h(x, u)}{\partial u} \right|_{(x_e, u_e)} = \begin{bmatrix} \frac{\partial h_1}{\partial u_1} & \dots & \frac{\partial h_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial u_1} & \dots & \frac{\partial h_p}{\partial u_m} \end{bmatrix}_{(x_e, u_e)} \in \mathbb{R}^{p \times m}$$

Given this linearized system, as we have already seen when we talked about stability, the Hartman-Grobman theorem states that if the linearized system is closed-loop BIBO stable, then the non-linear system is locally stable around the equilibrium point.

Note: We don't know how large this region of stability is, but we know that it exists.

To conclude, if we want to design controllers for non-linear systems, we can follow these steps:

1. Design a linear compensator for the linear model.
2. If the linear system is closed-loop stable, so will be the non-linear system in a neighborhood of the equilibrium.
3. Check in a non-linear simulation the robustness of your design with respect to typical deviations.